



Society of Petroleum Engineers

SPE-227043-MS

Novel Method to Drift and Pressure Test a Casing String without Intervention

H. Fethiza Tedjani, Saudi Aramco, Saudi Arabia; N. Albaijan, Baker Hughes Co., Saudi Arabia; A. Gundersen, Baker Hughes Co., United States; H. Al-Jarodi, Baker Hughes Co., Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227043-MS](https://doi.org/10.2118/227043-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper describes the first practical application, to the authors' knowledge, of using longstring casing wiper plugs to interventionlessly drift, wipe and pressure test a casing string during installation, and then subsequently cementing the casing string in place.

While installing certain critical wells the completion casing string must be examined for leaks and internal obstructions that could compromise future operations. Up until now the casing string has been drifted and pressure tested during installation by using multiple slickline and retrievable bridge plugs runs. This current practice has certain limitations, primarily the capability of the slick line, which is negatively affected by high mud weight, the slick line maximum over-pull load rating, and the well trajectory.

This was a 17,986 ft long/deep vertical well with a bottom hole temperature of 340°F. This new system eliminated the need for multiple slickline, and retrievable bridge plug runs, reduced the entire shoetrack length to less than 6 feet, and included a new type of high-performance one-way check valve in the float equipment. The new system saved more than 24 hours of rig time compared to the previous method. Three rupture disk longstring wiper plugs were used, each having drift gauge outer diameter to drift and pressure test the casing to 5,000 psi in three separate intervals. A fourth wiper plug with solid core was dropped behind the cement to wipe the casing clean during the cement job. All four longstring wiper plugs landed (bumped) as expected in the integrated shoetrack tool at the bottom of the casing string. The integrated shoetrack tool had four one-way backpressure valves, each rated to 15,000 psi differential backpressure qualified according to API-10F. Once the last wiper plug landed (bumped), 893 bbls of fluid had been circulated through the integrated shoetrack tool, and the floats were confirmed to hold.

This paper presents a new and novel method to drift, and pressure test a casing string using rupture disk longstring wiper plugs to drift and wipe the casing. The plugs were circulated down the casing string until they land at the end of the completion string, into a profile in a unique integrated shoetrack tool, at which point the casing string was pressure tested. After achieving a successful pressure test, pressure was further increased until the rupture disk in the rupture disk longstring wiper plug burst, at which point circulation was restored. Once the entire casing string had been run in hole and it had been drifted and pressure tested, the cement job was performed. The cement was displaced by a solid core longstring wiper plug. The solid

core wiper plug latched and sealed on the last previous longstring rupture disk wiper plug at the end of the cement displacement operation.

Introduction

During completion activities on oil and gas wells, such as those intended for future hydraulic fracturing operations, it is essential to inspect the completion pipe (casing or tubing) for leaks and internal obstructions that could jeopardize subsequent procedures. Traditionally this inspection has been performed by deploying a drift gage on wireline to drift the completion pipe (string), and a bridge plug on wireline to pressure test the completion string. However, there are constraints with this traditional method. Unless the well is vertical or the inclination is manageable, there is a limit to how deep the drift and bridge plug can be deployed with slickline. Wireline with conveying assistance, such as electrical tractors, is a potential workaround, but comes with its own set of disadvantages. When using slickline, for sufficiently deviated (the actual inclination depends on many factors) or horizontal wells it is necessary to segment the completion string length into smaller segments which can be drifted and pressure tested individually using slickline. Although this is a straightforward solution it results in having to perform several drifting and pressure testing runs for long deviated wells, which adds complexity, time and expense to the operation, and reduces the number of wells a single rig can drill and complete over a certain fixed period (typically measured in wells per month).

Current Solution

The current solution performs the following three purposes:

1. Wiping
2. Drifting
3. Pressure testing

Using a combination of the following technologies:

1. Wiper plugs
2. Floating drift
3. Slickline deployed drift gauge
4. Slickline deployed retrievable bridge plug

Wiping

Wiping involves providing a mechanical separation between two fluid volumes, one on each side of the wiper plug (in front of and behind). This is only performed when wiper plugs are dropped during the cement job.

Drifting

Conventional Drift. Drifting involves carrying a gauge through the completion string. A gauge is an object with a certain pre-determined outer diameter, with (cylinder) or without (solid) an inner diameter. The pre-determined outer diameter of the gauge is equal to the required drift diameter of the completion string. The required drift diameter of the completion string depends on the size and weight of the casing or tubing it consists of. It's also possible to have a completion string that consists of different sizes of casing and/or tubing. If the gauge successfully passes through a section of the completion string it is said to have been "drifted", confirming that it is free of obstructions and ready for subsequent operations. If there is a restriction in the completion string which prevent the gauge from passing through the gauge will stop, indicating that there is a problem with the completion string that needs to be addressed and resolved. One of the most common causes of unexpected restrictions in a casing string made up of casing or tubing that was

drifted at or prior to being run in hole is over-torqued (over-tightened) casing connections. Over-torqued casing connections are connections that had more torque applied than what is specified by the manufacturer (what the connection was designed for). When casing or tubing connections are over-torqued, typically, the outer diameter increases and the inner diameter decreases, until the connection reaches the point of complete failure.

Floating Drift. A special type of drift used while running in hole with a completion string is called a floating drift. The floating drift drifts the casing while it is being run in hole. The assembly has an outer diameter that is equal to the drift of the specific type of casing that is being run. The floating drift is dropped into the casing at the beginning of the operation. When the pipe is filled with fluid it rises to the surface due to buoyancy. The personnel on the rig floor can easily visually confirm that it has risen to the top and above the fluid level (out of the fluid) because it has a bright shining light on the top. If the floating drift doesn't float up to the surface, that is an indication that it identified a casing joint that doesn't meet the required drift. At that point the casing will be pulled out of hole until the floating drift is found, and the problematic casing joint will be removed and laid down.

Pressure Testing

Pressure testing involves providing a means to apply pressure at surface (surface pressure) to increase the internal pressure in the completion string. A pressure drop below a pre-determined acceptance criteria, for example 5%, over the duration of the pressure test, for example 15 min, verifies the integrity of the completion string.

The completion string is open at the bottom, at the shoetrack. The shoetrack of an oil and gas well refers to the lowermost section of the completion string, extending from the bottom of the float shoe up to the top of the landing collar—or to the top of the float collar if a landing collar is not present (Gundersen, 2023). The system is designed to allow fluid to exit the casing string into the annulus while preventing any backflow from the annulus into the casing. This one-way flow control is achieved through multiple redundant check valves installed in the shoetrack, which permit flow only from the casing to the annulus. These check valves also prevent cement from flowing back into the casing at the end of the cementing operation—a phenomenon known as U-tubing. It is necessary to allow fluid to exit the casing string into the annulus to be able to circulate through the completion string while it is run in hole, which is important for both well control (HSE) and to wash through any possible obstructions in the previous completion string or open hole.

However, because the completion string is open at the bottom, it is necessary to create a temporary plug/seal to perform the pressure test, before the completion string can continue into the well. This temporary plug is created by running into the well with a retrievable bridge plug on slickline. After the bridge plug is set the slickline can be left in hole, if the lubricator is rated to withstand the casing test pressure, otherwise the slickline must be pulled out of hole before the pressure test can be conducted. The lubricator is a long, high-pressure pipe attached to the top of a wellhead or Christmas tree, designed to allow the safe insertion of tools into a high-pressure well. The upper section of the lubricator assembly features a high-pressure grease injection system and sealing components to maintain well control by sealing around the slickline.

After a successful pressure test has been achieved, the bridge plug needs to be retrieved. If the slickline is already in the well at the right depth the crew can immediately start pulling out of hole once the bridge plug is unset. If the slickline was pulled out of the hole before the pressure test, it needs to be run back in hole down to the depth of the retrievable bridge plug, reattach to it, unset and then pull out of hole. In either case, once the bridge plug has been pulled out of hole the rig can continue running in hole with the completion string until it's time for the next drift and pressure test.

New Solution

To the authors' knowledge, the first publication describing an alternative method to the current solution using wiper plugs to drift and pressure test the completion string in a single operational effort, is the following US patent (Tedjani, 2020):

US 11,248,439 Plugs and Related Methods of Performing Completion Operations in Oil and Gas Applications, Applicant: Saudi Arabian Oil Company, Inventor: Hamza Fethiza Tedjani, filed April 30th, 2020.

It describes a method for pumping one or more wiper plugs from the surface down the completion string to perform the three previously described purposes (wiping, drifting and pressure testing) within a single operational effort. This eliminates the need for the following technologies currently used:

1. Floating drift
2. Slickline deployed drift gauge
3. Slickline deployed retrievable bridge plug

Because there is no well deviation (inclination and/or azimuth) or length limitation on pumping wiper plugs down the completion string, this new solution enables drifting and pressure testing the entire completion string in a single operation even in very long and highly deviated or horizontal wells. Additionally, removing the shoe track poses little to no negative impact, while offering clear advantages in terms of time and cost savings (Han, 2020). The next section will describe the technology that enables the new solution in more detail.

Technology Overview

The technology that enables the new solution consists of three different compatible components that are used together as a single system:

1. Integrated Shoetrack Tool
2. Longstring Wiper Plug (bottom plug).
3. Longstring Wiper Plug (top plug)

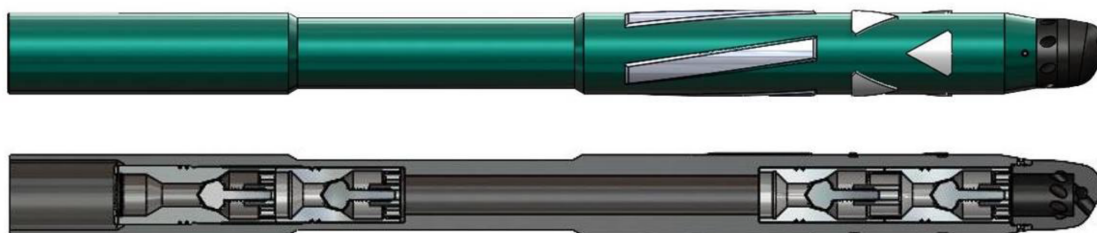


Figure 1—Integrated shoetrack tool. 4x backpressure valves. Qualified per API 10F to D24, R10, T400, P15. 360° coverage with crushed carbide diamonds (2x staggered rows), and left-hand spiral carbide blades. (a) Top: External view. (b) Bottom: Half cut view.



Figure 2—Integrated shoetrack tool. External view of bottom end. 360° coverage with crushed carbide diamonds (2x staggered rows), and left-hand spiral carbide blades.

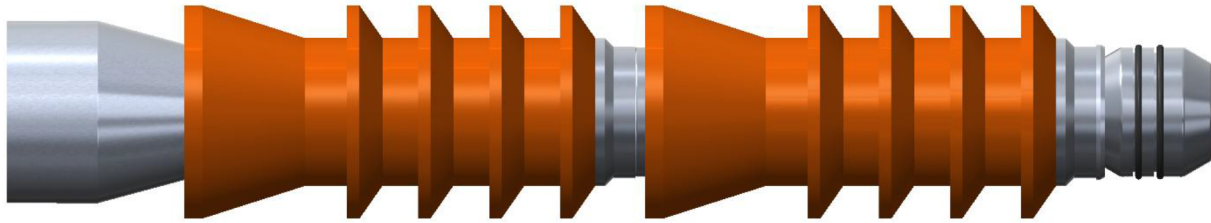


Figure 3—Longstring Wiper Plug. With rupture disk (bottom plug). Orange elastomer.



Figure 4—Longstring Wiper Plug. Solid nose (top plug). Black elastomer.

The same list expanded with additional details on each component:

1. Integrated Shoetrack Tool:
 - a. Wiper Plug Latch
 - i. 1x wiper plug mechanical-locking latch-in-profile.
 - b. Backpressure Valves
 - i. 4x manual-fill one-way check valves qualified to API 10F 4th edition "D24, R10, T400, P15".
 - c. Reaming and Wellbore Centralization
 - i. 360° coverage diamond shaped crushed carbide reamer structure, enabling reaming with or without rotation (reciprocation and/or rotation).
 - ii. Multiple left-hand (anti-aggressive) spiral blades centralizes the nose and scrapes away any obstructions from inside the previous casing string or from the inside of the open hole, with or without rotating the completion string from surface.
 - d. Wellbore Washing/Cleaning
 - i. The structure of the crushed carbide triangles and left-hand spiral blades generate turbulent flow while pumping, even without rotation, and this turbulence is further boosted (increased) when rotating the completion string. The increased turbulence improves the ability to lift and remove solids and cuttings.
 - e. Nose and Nose Jets
 - i. Eccentric guide nose
 - i. The nose (shown in black) purposefully has a smaller OD than the body where the crushed carbide triangles and blades sit. This allows the nose to more easily guide the casing string through obstructions without getting hung up. The transition from the smaller nose to the body where the crushed carbide triangles and blades sit has a straight taper (flat angle), which allows swaging the formation open using set down weight.
 - ii. Main Port

- i. Center located and downward directed washes and lifts cuttings.
- iii. Multiple Upjets
 - i. Accelerates the fluid velocity, and consequentially solids and cuttings, upwards in the annulus.
- 2. Longstring Wiper Plug (bottom plug):
 - a. With 5,000 psi rupture disk
 - i. Nominal burst pressure at 250°F, +/- 10% tolerance.
 - ii. Used for 3,000 psi casing integrity pressure test.
 - b. Orange elastomer, the industry standard for bottom plugs.
- 3. Longstring Wiper Plug (top plug):
 - a. Solid nose (no inner diameter / throughbore).
 - b. Black elastomer, the industry standard for top plugs.

Previous Industry Experience

Integrated shoetrack tools (IST) become popular several years ago in North America Land. They arose with the introduction of intentionally uncemented shoetracks. On intentionally uncemented shoetracks it's beneficial and desirable to have as short of a shoetrack as possible, which integrated shoetrack tools provide (Gundersen, 2023). Intentionally uncemented shoetrack started gaining in popularity around the late 2000s/early 2010s, both for liners and longstrings, and over the past decade there have been more than 1,000 wells completed with intentionally uncemented shoetrack on liners alone (Gundersen, 2023). The number of wells completed with intentionally uncemented shoetrack on longstring is likely several times larger. Despite this extensive industry experience this paper describes, to the authors' knowledge, the first worldwide application of drifting wiper plugs with high-pressure rupture disks to drift and pressure test the completion string while running in hole with casing.

Pre-Installation Validation

To reduce the technical risk associated with the field installation the technology was tested in a lab environment and in the workshop. Both the lab API 10F qualification test and the workshop plug bump and rupture disk burst test were successful.

Lab Test. While installing and cementing the casing string the shoetrack is required to allow flow from above (regular circulation/pumping) while preventing inflow from below (U-tubing). The American Petroleum Institute (API) Specification 10F "Cementing Float Equipment Testing" is the relevant industry specification for this type of equipment. It outlines how to test and rate the most common types of float equipment.

There are 4 different types of performance criteria for manual fill (non-auto fill) float equipment:

1. Flow durability
2. Forward flow rate
3. Temperature
4. Pressure

There are 5 different types of performance criteria for auto fill float equipment:

1. Flow durability
2. Forward flow rate

3. Temperature
4. Pressure
5. Reverse flow duration

The testing was performed according to the most up to date version of API 10F, which was the 4th Edition (published on July 2018, ERRATA 1 on June 2020, and ERRATA 2 on June 2020). The float equipment was manual fill (non-auto fill).

Dynamic Flow Test

- 24 hours of circulation at a flow rate of 10 bpm.

Static High-Pressure & High-Temperature Test

- 15,000 psi backpressure at 400°F.

The float equipment was qualified to D24, R10, T400, P15. D24 and R10 represents 24 hours of circulation at 10 bpm during the *Dynamic Flow Test* part of the qualification test. T400 and P15 represents 400°F temperature and 15,000 psi differential backpressure (from below to above) during the *Static High-Pressure & High-Temperature Test* part of the qualification test. The float valves passed all parts of the qualification test.

Workshop Function Test. After the equipment arrived at the workshop, a test was performed. This pre-installation trial test, "System Integrity Test", consisted of a plug bump and wiper plug rupture disk rupture/burst test at ambient temperature. The burst pressure of the metal rupture disk decreases with increasing temperature. It has a nominal burst at 5,000 psi at 250°F. Since the test was conducted in the workshop at ambient temperature the nominal burst was higher, 5,353 psi to be specific. The design tolerance is +/-10% of nominal. Thus, the burst range at ambient was 4,818-5,888 psi. The plug, with the mechanical latch profile removed, was installed into the latch landing profile, so that could be easily removed after the test. Then the system was pressured up from above. It was confirmed that the system held 3,000 psi differential pressure. Then pressure was increased until the 5,000 psi rupture disk in the wiper plug burst at 5,630 psi, 5.2% (5,630 /5,353) over nominal, well within the +/- 10% tolerance. The test parameters and results are summarized in [Table 1](#).

Table 1—Workshop function test.

#	Parameter	Value
1	Rupture disk burst pressure tolerance	+/- 10%
2	Nominal rupture disk burst pressure @ 250°F	5,000 psi
3	Minimum rupture disk burst pressure @ ambient	4,818 psi
4	Nominal rupture disk burst pressure @ ambient	5,353 psi
5	Maximum rupture disk burst pressure @ ambient	5,888 psi
6	System pressure test	3,000 psi
7	Actual rupture disk burst pressure @ ambient	5,630 psi
8	Actual rupture disk burst pressure @ ambient compared to nominal @ ambient	5.2% (5,630 /5,353)
9	The 3,000 psi pressure test was successful?	Yes
10	The rupture disk burst within the stated tolerance?	Yes
11	The rupture disk burst within tolerance?	Yes

Field Installation

The field installation occurred in Saudi Arabia in August 2024. The new technology equipment used for the installation was 1 ea integrated shoetrack tool, 3 ea wiper plugs with 5,000 psi rupture disk and drift gauge OD, and 1 ea wiper plug with solid core (no rupture disk). These details are summarized in [Table 2](#).

Table 2—New technology equipment used.

#	Description	Quantity
1	4.5" Integrated Shoetrack Tool	1
2	Longstring Wiper Plug (bottom plug) with 5,000 psi rupture disk and 3.701" in gauge outer diameter (OD)	3
3	Longstring Wiper Plug (top plug)	1

Table 3—Installation details.

Parameter	Value
Country	Saudi Arabia
True Vertical Depth (TVD)	17,986 ft
Deviation/Inclination	Vertical
Bottom Hole Temperature (BHT)	340°F
Completion string	4.5" 15.2# TenarisHydril (TSH) Blue TN-125HC
Drift size for 4.5" casing string	3.701"

The well was a very deep vertical well, with a True Vertical Depth of 17,986 ft. The bottom hole temperature was 340°F. These details are summarized in [Table 4](#).

Table 4—Installation Success Criteria.

#	Description
1	Successfully run to target depth without operational issues related to the integrated shoetrack tool.
2	Successfully launch and displace the required number of drifting longstring wiper plugs with rupture disk to latch into the integrated shoetrack tool and perform a 3,000-psi pressure test against it without a leak from the wiper plug.
3	Successfully rupture the rupture disks on the drifting longstring wiper plugs within accepted range and regain circulation afterwards.
4	Successfully perform cement job with the integrated shoetrack tool.
5	Successfully launch and displace the solid core longstring wiper plug behind cement to latch into the last drifting longstring wiper plug on the integrated shoetrack tool.
6	Solid core longstring wiper plug successfully holds bump pressure after it latches into the last drifting longstring wiper plug.
7	Integrated shoetrack tool successfully holds back pressure (prevents back flow) throughout the job.

The success criteria used to evaluate the result of the field installation are shown in [Table 3](#). All 7 success criteria were met by the installation.

A detailed list of the operational steps is included in Appendix A: Table A-1. In summary, the open hole was prepared then the rig began running in hole with casing. After 148 casing joints had been picked up, the first wiper plug with 5,000 psi rupture disk and drift gauge OD was picked up. These wiper plugs had been pre-loaded into short pup-joints (casing joints shorter than the regular API lengths). The pup joint was made up to the casing string and then the wiper plug was pumped down the casing until it landed ("bumped") in the latch-in-profile of the integrated shoetrack at the bottom of the completion string, after having successfully drifted all the casing joints. Then a 15 min 5,000 psi pressure test was performed. With a successful pressure

test, pressure was increased until circulation was regained. The rig continued to run in hole with casing joints. After 296 casing joints had been picked up, the process with the wiper plug was repeated and another pressure test performed. This was performed a third time after 441 casing joints had been picked up. After the third and last pressure test had been completed, the cement job was performed. After all the cement had been pumped, a 4th wiper plug was dropped. This was a solid core wiper plug without a rupture disk and without a drift gauge OD. This wiper plug was displaced until it landed into the tail of the 3rd wiper plug, confirmed with a pressure-tight seal upon plug latch ("bump").

Results

Pump rates and pressures relevant for drifting and pressure test are included in [Table 5](#). Wellbore volumes relevant for wiper plug wiping efficiency are shown in [Table 6](#).

Table 5—Field installation results. Volumes, pump rates and pressures relevant for drifting and pressure test.

Section	Casing interval #1	Casing interval #2	Casing interval #3
Casing joints	#148	#296	#441
Displacement pressure with	3 BPM @ ~450 psi 5 BPM @ ~ 1050 psi	3 BPM @ ~600 psi 5 BPM @ ~ 1,400 psi	3 BPM @ ~1,800 psi 5 BPM @ ~ 2,500 psi
Calculated (theoretical) displacement volume	85 bbls	171 bbls	256 bbls
Actual displacement volume (plug landed/bumped with)	86 bbls	174 bbls	265 bbls
15 min pressure test successful	Yes	Yes	Yes
Pressure test completed with	3,150 psi	3,080 psi	3,010 psi
Rupture disk rupture at	5,400 psi	5,250 psi	5,400 psi
Regain circulation with	3 BPM @ ~515 psi	2 BPM @ ~300 psi	3 BPM @ ~1,450 psi

Table 6—Field installation results. Wellbore volumes relevant for wiper plug wiping efficiency.

Min Casing ID (in)			3.751			
Nom Casing ID (in)			3.826			
Max Casing ID (in)			3.904			
Wiper Plug #			1	2	3	4
Wiper Plug Type			Rupture Disk	Rupture Disk	Rupture Disk	Solid
Total Casing Length (ft)			6,002	11,990	17,985	17,985
Volumes (bbls)	Calculated	Min Casing ID	82	164	246	246
		Nom Casing ID	85	171	256	256
		Max Casing ID	89	178	266	266
	Actual		86	174	265	255
	Variance between calculated and actual		1	3	9	-1
			1.2%	1.8%	3.5%	-0.4%

Lessons Learned

As expected ahead of time, it was difficult to pre-load the wiper plugs into the casing pup joints at the workshop prior to the job. It was achieved using equipment that is used to "stab" liner hanger running tools equipment (the equipment that is pulled out of the hole after the job) into liner hanger equipment (the

equipment that stays downhole after the job). Long term, there would be benefits to both pre-job preparation and on-the-rig operation in having a cement head that can accommodate the longer than standard wiper plugs used on this installation.

Conclusion

While installing certain critical wells the completion casing string must be examined for leaks and internal obstructions that could compromise future operations. Up until now the casing string has been drifted and pressure tested during installation by using multiple slickline and retrievable bridge plug runs. This current practice has certain limitations, primarily the capability of the slick line, which is negatively affected by high mud weight, the slick line maximum over-pull load rating, and the well trajectory.

This paper has described the first practical application, to the authors' knowledge, of using longstring casing wiper plugs to interventionlessly drift, wipe and pressure test a casing string during installation, and then subsequently cementing the casing string in place. The plugs were circulated down the casing string until they landed at the end of the completion string, into a profile in a unique integrated shoetrack tool, at which point the casing string was pressure tested. After achieving a successful pressure test, pressure was further increased until the rupture disk in the rupture disk longstring wiper plug burst, at which point circulation was restored. Once the entire casing string had been run in hole and it had been drifted and pressure tested, the cement job was performed. The cement was displaced by a solid core longstring wiper plug. The solid core wiper plug latched and sealed in the last previous longstring rupture disk wiper plug at the end of the cement displacement operation.

Three rupture disk longstring wiper plugs were used, each having drift gauge outer diameter to drift and pressure test the casing to 5,000 psi in three separate intervals. The drift and pressure test operations were performed with the following casing lengths run in hole: 1st: 6,002 ft (148 joints), 2nd: 11,990 ft (296 joints), and 3rd: 17,985 ft (441 joints). A fourth wiper plug with solid core was dropped behind the cement to wipe the casing clean during the cement job.

All four longstring wiper plugs landed (bumped) as expected in the integrated shoetrack tool at the bottom of the casing string. The integrated shoetrack tool had four one-way backpressure valves, each rated to 15,000 psi differential backpressure qualified according to API-10F. Once the last wiper plug landed (bumped), 893 bbls of fluid had been circulated through the integrated shoetrack tool, and the floats were confirmed to hold.

This was a 17,986 ft long/deep vertical well with a bottom hole temperature of 340°F. This new system eliminated the need for multiple slickline, and retrievable bridge plug runs, reduced the entire shoetrack length to less than 6 feet, and included a new type of high-performance one-way check valve in the float equipment. The new system saved more than 24 hours of rig time compared to the previous method.

Abbreviations

Shortened form Long form

°F	Degrees on the Fahrenheit temperature scale
API	The American Petroleum Institute (API)
API 10F	The American Petroleum Institute (API) Specification 10F "Cementing Float Equipment Testing"
bbl	Barrel
BPM	Barrel per minute
CHC	Circulate hole clean
ID	Inner diameter
in	Inches
IST	Integrated Shoetrack Tool

Max	Maximum
Min	Minimum
Nom	Nominal
OD	Outer diameter
POOH	Pull out of hole
RIH	Run in hole

References

1. Inventor, Tedjani, H. F. 2020. Plugs and Related Methods of Performing Completion Operations in Oil and Gas Applications. US Patent No. 11,248,439.
2. Han, Kyungnam, Alexeyenko, Alexey, Fethiza Tedjani, Hamza, and Maksim Oparin. "Case Study: Monobore Cemented Completion Without Shoe Track for Unconventional Gas Wells." Paper presented at the International Petroleum Technology Conference, Dhahran, Kingdom of Saudi Arabia, January 2020. doi: <https://doi.org/10.2523/IPTC-20222-Abstract>
3. Gundersen, Anders, Gomez, Ricardo, Kadler, Chelsea, Broussard, Taylor, and Arjang Gandomkar. "Intentional Wet Shoetrack Completions for Unconventional Oil and Gas Resources: Methods and Techniques." Paper presented at the SPE Oklahoma City Oil and Gas Symposium, Oklahoma City, Oklahoma, USA, April 2023. doi: <https://doi.org/10.2118/213081-MS>

Appendix A

Table A-1—Operational steps.

Step	Description
1	Open hole (OH) logging run.
2	Perform wiper trip with drilling/reaming assembly.
3	Circulate Hole Clean.
4	Pull out of hole with drilling/reaming assembly.
5	Test Blow Out Preventer.
6	Rig up casing handling equipment, casing tongs and Casing Running Tool.
7	Install 4.5" Integrated Shoetrack Tool with Drifting and Pressure Testing Wiper Plugs.
8	Verify float valves are holding. Floats are holding.
9	Install floating drift with 3.701" in gauge OD in casing string.
10	Run in hole with casing.
11	Fill up every casing joint.
12	At casing joint #148, retrieve the floating drift and break circulation. Record the standpipe pressure.
13	Pick up and make up pup joint #1 with pre-loaded drifting and pressure testing plug and casing joint #149 and prepare the surface lines for the displacement.
14	Flush lines then pressure test them to 8,000 psi from cement unit for 10 min, confirm good test.
15	Start displacing the drifting plug with 5 BPM for 70 bbls @ 1080 psi, then reduce the pump rate to 3 BPM @ 500 psi prior bumping the plug with 85 bbls theoretical volume. Plug bumped with 86 bbls. Bleed off and switched to cement unit.
16	Pressure test casing from cement unit with 3,000 psi for 15 min, Good test. (start test: 3125 psi - End Test: 3197 psi). Increase the pressure to rupture the rupture disk. Disk burst @ 5,400 psi. Regain circulation with 3 bpm @ 470 psi. R/D surface line and lay down pup joint.
17	Install floating drift "3.701 in" and then count RIH w/ 296 "Total" joints of 14-1/2" 15.2# T. Blue TN-125HC. fill up every joint.
18	With casing joint #296, retrieve floating drift and break circulation to recorded pressure.
19	Pick up and make up pup joint #2 with pre-loaded drifting and pressure testing plug and casing joint #297 and prepare the surface lines for the displacement.
20	Start displacing the drifting plug with 5 BPM for 155 bbls @ 1300 psi, then reduce the pump rate to 3 BPM @ 600 psi prior bumping the plug with 171 bbls theoretical volume. Plug bumped with 173 bbls. Bleed off and switched to cement unit.

Step	Description
21	Pressure test casing from cement unit with 3,000 psi for 15 min, Good test. (start test: 3080 psi - End Test: 3,280 psi). Increase the pressure to rupture the rupture disk. Disk burst @ 5,250 psi. Regain circulation with 2 bpm @ 300 psi. R/D surface line and lay down pup joint.
22	Install floating drift "3.701 in" and then count run in hole w/ 441 "Total" joints of 14-1/2" 15.2# T. Blue TN-125HC. fill up every joint.
23	Pick up and make up pup joint #3 with pre-loaded drifting and pressure testing plug and casing joint #442 & 443 (Drifted with floating drift) then Pick up tubing hanger and landed it as per FMC Eng.
24	Pressure test the tubing hanger and meanwhile prepare surface lines with cement head
25	Start displacing the drifting plug with 5 BPM for 240 bbls @ 2,500 psi, then reduce the pump rate to 3 BPM @ 1,800 psi prior bumping the plug with 256 bbls theoretical volume. Plug bumped with 265 bbls. Bleed off and switched to cement unit.
26	Pressure test casing from cement unit with 3,000 psi for 15 min, Good test. (start test: 3,000 psi - End Test: 3,330 psi). Increase the pressure to rupture the rupture disk. Disk burst @ 5,400 psi. Regain circulation with 3 bpm @ 1,800 psi.
27	Circ bottom up and HPJSM prior the cement job.
28	Start cement job as per the following: • 150 bbls of 98 pcf low rheology mud. • 120 bbls of 106 pcf mud push spacer • 98 bbls of 118 pcf cement slurry
29	Breck cement head, flush line, drop Baker Hughes Solid Core Long String Plug and displaced cement as following: • 20 bbls of 106 pcf mud push spacer. • 235 bbls of 98 pcf low mud.
30	Bump Plug with 3 BPM @ final displacement pressure of 2,100 psi. Increase the pressure to 3,000 psi and hold it for 15 min. Good test. Bleed of to check floats. Floats are holding.
31	Installation completed successfully.